```
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt

# Load the image
img_path = '/content/F1 🏁🏎️ .#f1 #aesthetic #formula1 #racing (00.jpg'
img = Image.open(img_path)

# Convert to numpy array
img_array = np.array(img)

# Perform a numpy operation: Flip the image horizontally
flipped_array = np.flip(img_array, axis=1)

# Display the original and the modified image
fig, axes = plt.subplots(1, 2, figsize=(12, 6))
axes[0].imshow(img_array)
axes[0].set_title('Original')
axes[0].axis('off')

axes[1].imshow(flipped_array)
axes[1].set_title('Flipped (Numpy)')
axes[1].axis('off')

plt.show()
```

```
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img_path = '/content/F1 🏁🏎️ .#f1 #aesthetic #formula1 #racing (00.jpg'
img = Image.open(img_path)

arr = np.array(img)
print("Shape", arr.shape)
print("Data Type", arr.dtype)
```
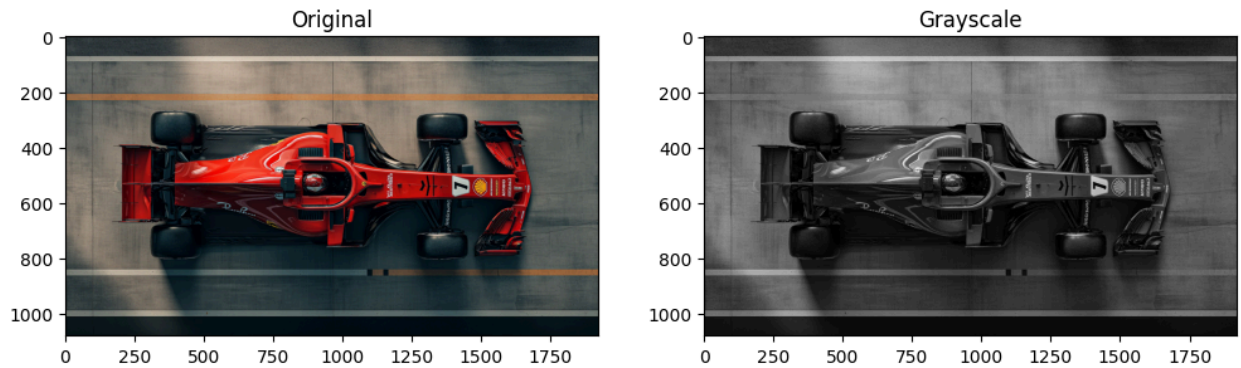
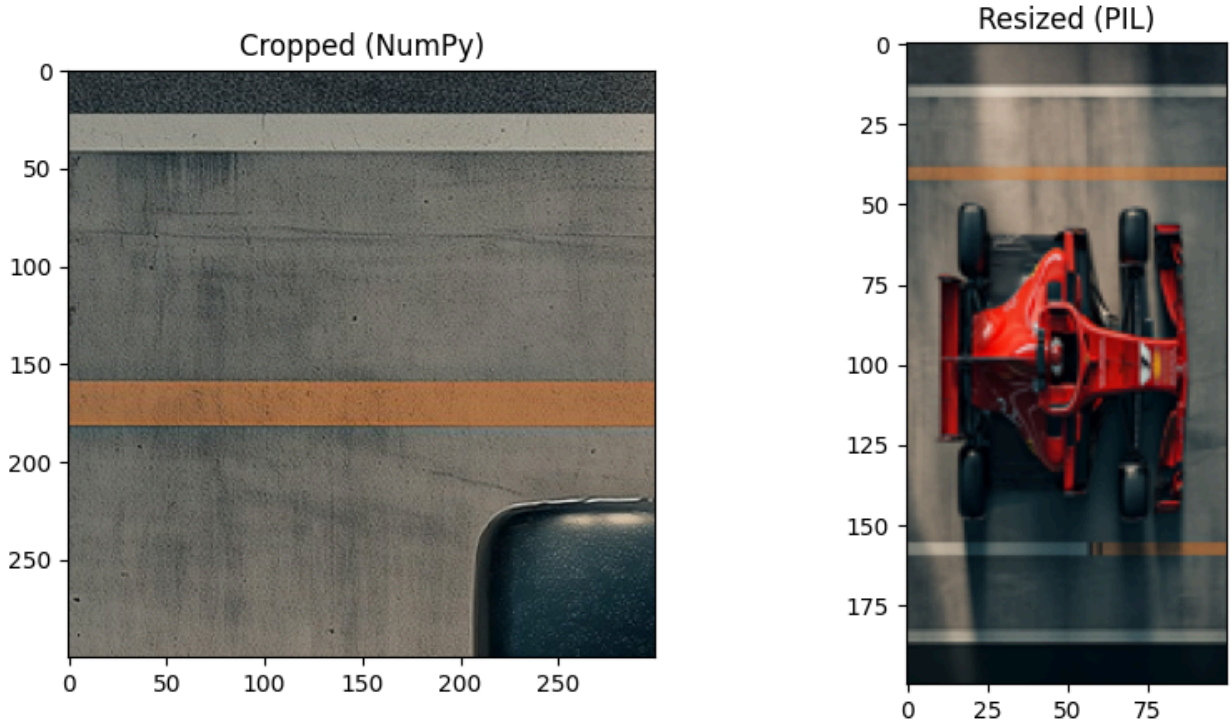

```
Shape (1080, 1920, 3)
Data Type uint8
```

```
gray = np.mean(arr, axis=2).astype(np.uint8)
```

```
gray = np.mean(arr, axis=2).astype(np.uint8)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1); plt.title('Original'); plt.imshow(arr)


plt.subplot(1, 2, 2); plt.title('Grayscale');
plt.imshow(gray, cmap='gray')
plt.show()
```

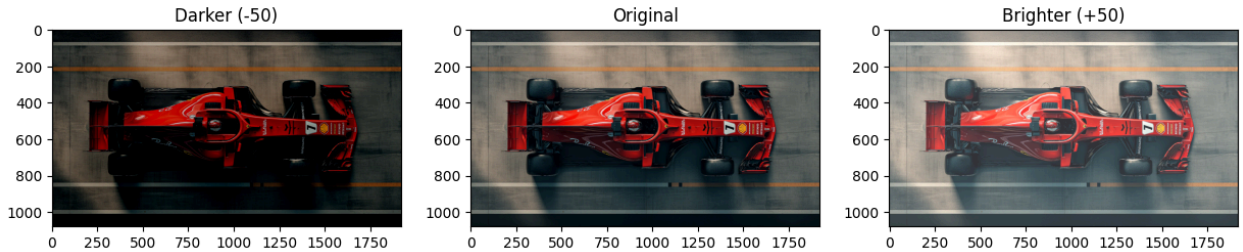



```
# Fix: Ensure the slicing indices are valid (start < end)
crop = arr[50:350, 100:400]
resized = np.array(img.resize((100, 200)))

plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.title('Cropped')
plt.imshow(crop)

plt.subplot(1, 2, 2)
plt.title('Resized')
plt.imshow(resized)
plt.show()
```



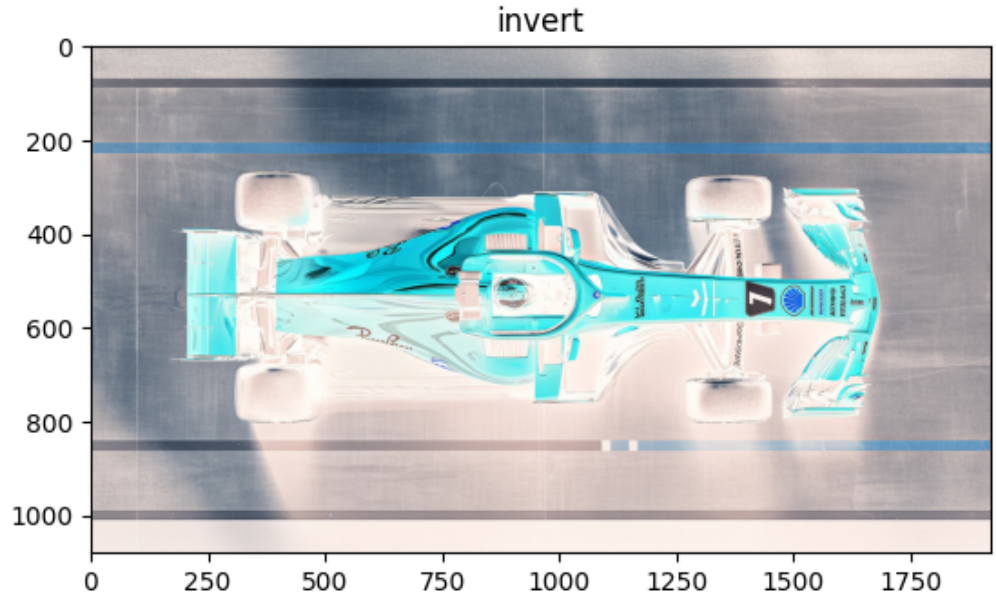
Start coding or generate with AI.

```
brightness_offset = 50

bright_arr = np.clip(arr.astype(np.int16) + brightness_offset, 0, 255).astype(np.uint8)
dark_arr = np.clip(arr.astype(np.int16) - brightness_offset, 0, 255).astype(np.uint8)

plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1); plt.title('Darker (-50)'); plt.imshow(dark_arr)
plt.subplot(1, 3, 2); plt.title('Original'); plt.imshow(arr)
plt.subplot(1, 3, 3); plt.title('Brighter (+50)'); plt.imshow(bright_arr)

plt.show()
```



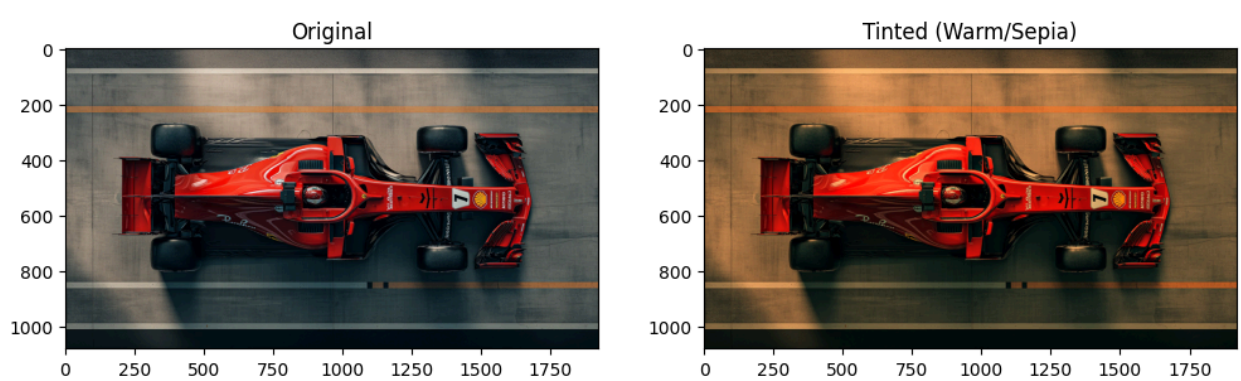

```
invert = 255 - arr
plt.subplot(1, 1, 1); plt.title('invert'); plt.imshow(invert)
```

```
<matplotlib.image.AxesImage at 0x7ed27951f4d0>
```



Start coding or generate with AI.

```
# Define a tint color (e.g., warm/orange tint)
# R, G, B scaling factors
tint_factor = np.array([1.2, 0.9, 0.6])

# Apply the tint by multiplying and then clipping to stay in 0-255 range
tinted_arr = np.clip(arr * tint_factor, 0, 255).astype(np.uint8)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.title('Original')
plt.imshow(arr)

plt.subplot(1, 2, 2)
plt.title('Tinted (Warm/Sepia)')
plt.imshow(tinted_arr)

plt.show()
```



Double-click (or enter) to edit